

Friendly Order Type Through Incomparability Components

A finite structural case of an open problem,
with Cartesian products and transfinite extensions

For every finite poset P ,

$$o_{\perp}(P) = |P| - c_{\perp}(P) \quad \text{and hence} \quad w(M^r(P)) = \omega^{|P| - c_{\perp}(P)}.$$

Here $c_{\perp}(P)$ counts the connected components of the incomparability graph, including isolated vertices. The first quantity is also the rank of that graph's graphic matroid.

Abstract

Vialard introduced friendly order type to compute the ordinal width of the Dershowitz–Manna multiset ordering and asked for further structural and compositional descriptions. We study the finite structural case. We prove that the friendly order type of a finite poset is its number of elements minus its number of incomparability components. One proof bounds every open-ended bad sequence by a graph-forest potential; the matching lower bound constructs connected linear extensions of the components. This identifies the invariant with a familiar graph rank and gives polynomial-time value and witness algorithms.

We derive exact formulas for finite lexicographic substitutions and Cartesian products. We also prove additivity over arbitrary well-ordered ordinal sums, which yields a transfinite evaluation theorem for all well-partial-orders with finite incomparability components. Explicit countable examples show that neither graph-isomorphism invariance nor a simple connected-component correction to maximal order type extends to arbitrary infinite bases. Independent residual-rank computations verify the finite formula on all 101,660 naturally labelled posets of sizes at most seven; additional product and substitution checks are supplied as executable Python.

Research status. The proofs below are complete mathematical arguments, but have not been externally refereed or proof-assistant checked. The result solves a specified finite subproblem and an infinite extension, not the entire published research program. A targeted literature search did not locate the component formula; that search does not establish priority. The general multiset-width theorem is Vialard's, not a new claim of this manuscript.

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1 The documented problem and the scope of the answer

In the conclusion of *Ordinal Measures of the Set of Finite Multisets*, Vialard asks about the friendly order type: “How does it relate to other concepts? Can it be computed compositionally for more operations?” [1, p. 87:11]. The 2024 doctoral thesis again asks for compositional computation and a deeper interpretation of this invariant [2, Conclusion, pp. 109–110]. These are open-ended research questions rather than one numbered yes/no conjecture.

The concrete problem selected here is the following finite structural specialization of that program.

Problem 1.1 (Finite structural subproblem). Find an exact structural expression for $\mathbf{o}_\perp(P)$ for every finite poset P , relate it to a familiar invariant, and use that expression to compute it for additional poset constructions. Determine how far the expression extends to infinite well-partial-orders.

The literature consulted does not separately designate Problem 1.1 as a named open conjecture. We therefore do not claim to have settled a previously named conjecture. The contribution is a proposed structural solution to a concrete part of the documented program. The finite case is not merely a question about finite ordinal widths: $M^r(P)$ is usually infinite even when P is finite, and its width is the transfinite ordinal $\omega^{\mathbf{o}_\perp(P)}$.

The principal statements are as follows.

Domain	Exact conclusion
Finite P	$\mathbf{o}_\perp(P) = P - c_\perp(P)$; equivalently, graphic-matroid rank.
Finite substitutions	Each nontrivial index component merges all its fibers into one incomparability component.
Finite $A \times B$, $ A , B \geq 2$	$\mathbf{o}_\perp(A \times B) = A B - 1 - b - t$, where b, t indicate the existence of a least and greatest element in the product.
Well-ordered sums	$\mathbf{o}_\perp(\sum_{i < \delta} P_i) = \sum_{i < \delta} \mathbf{o}_\perp(P_i)$, using ordinary ordinal addition.
Finite-component wpos	The preceding sum becomes $\sum_{i < \delta} (C_i - 1)$ over the ordered incomparability components.

The finite component theorem, its algorithmic form, the substitution formula, and the Cartesian formula are proved independently of the multiset-width theorem. The transfinite argument uses only the residual definition of friendly order type and well-founded tree ranks. The conversion from friendly order type to multiset width invokes Vialard’s theorem explicitly.

2 Definitions and conventions

2.1 Posets, residuals, and tree ranks

All orders in this article are partial orders, not merely quasi-orders. A *well-partial-order* (wpo) is a poset with no infinite bad sequence. A finite sequence $x_0, \dots, x_{k-1}$ in P is *bad* when

$$i < j < k \implies x_i \not\leq_P x_j.$$

Every finite poset is a wpo. For $x \in P$, set

$$\uparrow_P x = \{y \in P : x \leq_P y\}, \quad P_{\not\leq x} = P \setminus \uparrow_P x.$$

All subsets carry the induced order. After a bad sequence s , the residual is $P \setminus \bigcup_{x \in s} \uparrow_P x$. This is precisely the set of points that can still be appended to s while preserving badness.

For a well-founded rooted tree, a leaf has rank zero, and a node has rank

$$\sup\{\text{rank of a child} + 1 : \text{child of the node}\}.$$

The rank of the tree is the rank of its root. An empty supremum is zero. In particular, the tree consisting only of its empty sequence has rank zero. The tree $\text{Bad}(P)$ of finite bad sequences is well-founded for a wpo. We write $\text{o}(P)$ for its rank. This is the usual maximal-order-type invariant, but only its rank definition is needed in our transfinite proof. It satisfies

$$\text{o}(P) = \sup_{x \in P} (\text{o}(P_{\not\leq x}) + 1). \quad (1)$$

For a finite poset, $\text{o}(P) = |P|$: reverse any linear extension to obtain a bad sequence containing all elements.

2.2 Friendly order type

We use Vialard's definition of open-ended bad sequences [2, Definition 5.3.2]. Two points are *incomparable*, written $x \perp y$, when neither $x \leq y$ nor $y \leq x$ holds. Write $P_{\perp x} = \{y \in P : y \perp x\}$.

Definition 2.1 (Open-ended sequence and friendly order type). The empty sequence is open-ended. If s is open-ended and its current residual is R , then sx is open-ended when $x \in R$ has an incomparable point $y \in R$. Such a point y is called a *friend* of x at that step. Let $\text{Bad}_{\perp}(P)$ be the tree of open-ended bad sequences. Its rank is the *friendly order type* $\text{o}_{\perp}(P)$.

The friend need not be a member of the eventual selected sequence, and need not remain forever. It must exist in the residual *at the step where it is used*. Incomparability guarantees that it is not deleted by selecting x , because $x \not\leq y$.

The defining recurrence is

$$\text{o}_{\perp}(P) = \sup_{\substack{x \in P \\ P_{\perp x} \neq \emptyset}} (\text{o}_{\perp}(P_{\not\leq x}) + 1). \quad (2)$$

For finite P , this is a maximum, with value zero if no move is permitted, and the rank equals the maximum length of an open-ended bad sequence. In particular, $\text{o}_{\perp}(\emptyset) = 0$ and $\text{o}_{\perp}(P) = 0$ for every chain.

2.3 The graph and the multiset order

The *incomparability graph* $G_{\perp}(P)$ is the simple undirected graph with vertex set P and edges $\{x, y\}$ exactly when $x \perp y$. For finite P , let $c_{\perp}(P)$ be its number of connected components. Isolated vertices count as components, and $c_{\perp}(\emptyset) = 0$. For any finite graph G , write

$$r(G) = |V(G)| - c(G).$$

We will prove directly that this is both the maximum number of edges in a forest contained in G and the rank of an incidence matrix.

For completeness, the multiset ordering used here is not multiset embedding. For finite multisets m, n over P , cancel their common multiplicities and write $m' = m \setminus (m \cap n)$ and $n' = n \setminus (m \cap n)$. Then

$$m \leq_r n \iff m = n \text{ or } (\forall x \in m') (\exists y \in n') x <_P y. \quad (3)$$

No injective matching of the remaining occurrences is required. Write $M^r(P)$ for the resulting ordered set of finite multisets, including the empty multiset [1, Definition 1.6].

The ordinal width $w(Q)$ of a wpo Q is the rank of its tree of finite sequences of pairwise incomparable points. For finite Q it equals the ordinary maximum antichain size. For infinite Q it is generally not the supremum of the finite antichain sizes: the rank also records the nested structure of the possible continuations.

Theorem 2.2 (Vialard). *For every wpo P ,*

$$w(M^r(P)) = \omega^{\text{o}_\perp(P)}.$$

This is [1, Theorem 3.4]. We use it as a published theorem; we do not claim to reprove its general multiset argument. If P is empty, $M^r(P)$ has one element and the formula correctly gives $1 = \omega^0$.

3 The finite component theorem

Theorem 3.1 (Finite component formula). *For every finite poset P ,*

$$\boxed{\text{o}_\perp(P) = |P| - c_\perp(P)}.$$

Moreover, an open-ended bad sequence of this length can be constructed without evaluating the residual-rank recurrence.

The proof separates a graph-theoretic upper bound from an order-theoretic construction. The latter separation matters: connectivity alone does not make an arbitrary vertex ordering into a bad sequence.

3.1 Upper bound: every legal move consumes forest rank

Lemma 3.2 (Graph-rank decrease). *Let R be a finite poset, let $x \in R$ have a friend $y \in R$, and put $S = R \setminus \uparrow_R x$. Then*

$$r(G_\perp(S)) \leq r(G_\perp(R)) - 1.$$

Proof. A forest on a graph with v vertices and c connected components has at most $v - c$ edges: in each component, an acyclic graph on its vertices has at most one fewer edge than vertices. Choosing a spanning tree in each component attains $v - c$. Thus $r(G)$ is the maximum forest size.

Choose a spanning forest F of $G_\perp(S)$. Since $x \perp y$, the point y belongs to S , whereas x does not. Adding the new vertex x and the edge $\{x, y\}$ to F cannot create a cycle. It produces a forest in $G_\perp(R)$ with $r(G_\perp(S)) + 1$ edges. The asserted inequality follows. $\square$

Every step of an open-ended sequence decreases this nonnegative integer potential by at least one. Therefore

$$\text{o}_\perp(P) \leq r(G_\perp(P)) = |P| - c_\perp(P). \quad (4)$$

This proof allows an upset deletion to remove many points and to split old components into new ones. It does not assume that a move simply removes one vertex, or that connectivity is preserved after every possible move.

3.2 Incomparability components are ordinal-sum blocks

Lemma 3.3 (Ordered components). *Distinct connected components of $G_{\perp}(P)$ are uniformly comparable: for two such components C, D , either every point of C is below every point of D , or every point of D is below every point of C . These comparisons linearly order the components. This holds for arbitrary posets, finite or infinite.*

Proof. Points in different components must be comparable. Fix a point z outside a component C . If $u \perp v$ are adjacent vertices in C , the directions of the comparisons with z cannot differ: $u < z < v$ would imply $u < v$, and $v < z < u$ would imply $v < u$. Following a finite graph path shows that either all of C is below z or all of C is above z .

Now vary z along a path in a second component D ; the same argument shows that the direction is constant throughout D . This proves uniformity. Totality follows from comparability between components, and transitivity follows from transitivity in P . $\square$

For finite P we may consequently write

$$P = C_1 + C_2 + \cdots + C_k,$$

where $+$ is ordinal sum: every point of an earlier block is below every point of a later block. Each induced graph $G_{\perp}(C_i)$ is connected.

3.3 Connected linear extensions

Lemma 3.4 (A boundary point with a friend). *Let C be a finite poset whose incomparability graph is connected. If D is a nonempty proper order ideal of C , then some minimal point v of $C \setminus D$ is incomparable to a point of D .*

Proof. Suppose the assertion fails. Every minimal point v of $C \setminus D$ is then comparable to every $d \in D$. We cannot have $v \leq d$, because D is an order ideal and $v \notin D$. Thus $d < v$ for every such d, v .

Every point z of the finite poset $C \setminus D$ lies above a minimal point v of that poset. Hence $d < v \leq z$ for every $d \in D$. There are therefore no incomparability edges between the two nonempty sets D and $C \setminus D$, contradicting connectivity. $\square$

Lemma 3.5 (Connected-prefix extension). *If C is a finite poset with connected incomparability graph and $|C| = m$, there is a linear extension $v_1, \dots, v_m$ such that every v_j with $j \geq 2$ has an incomparable point among $v_1, \dots, v_{j-1}$. In particular, every nonempty prefix induces a connected incomparability graph.*

Proof. Start with any minimal point v_1 . Suppose an ideal $D = \{v_1, \dots, v_j\}$ has been constructed. If it is proper, Lemma 3.4 supplies a minimal remaining point v_{j+1} with a friend in D . Appending a minimal remaining point preserves the linear-extension property and keeps the selected set an ideal. Its friend connects the new point to the existing prefix. Continue until all points have been selected. The one-point case is included. $\square$

Corollary 3.6. *If $G_{\perp}(C)$ is connected and C is finite, then $\alpha_{\perp}(C) = |C| - 1$.*

Proof. Use Lemma 3.5 and read the extension backwards, omitting its first point:

$$v_m, v_{m-1}, \dots, v_2.$$

After the later points have been removed, v_j is maximal in the remaining prefix $\{v_1, \dots, v_j\}$. Its upset in that prefix is therefore the singleton $\{v_j\}$. It also has an incomparable friend in the earlier prefix. Every move is legal, giving length $m - 1$. The reverse inequality is (4). $\square$

3.4 Completion of the proof and an explicit certificate

Proof of Theorem 3.1. The empty case is immediate. Decompose a nonempty P into its ordered incomparability components $C_1 < \dots < C_k$. For each C_i , choose the extension of Lemma 3.5, and retain its reverse with the first point omitted. Process these lists in the order $C_k, C_{k-1}, \dots, C_1$.

While working in C_i , all points of earlier components are still present: selecting a point in a later component never removes a smaller point. The unprocessed prefix inside C_i is also still present. A chosen point has its designated friend in that prefix, and is maximal within that prefix. Its upset may delete leftover points of higher components, but this cannot affect its friend or any future move in a lower component. Thus the entire concatenated sequence is open-ended.

Its length is

$$\sum_{i=1}^k (|C_i| - 1) = |P| - k = |P| - c_{\perp}(P).$$

Together with (4), this proves the theorem. $\square$

Choose, for each v_j in a component extension with $j \geq 2$, one earlier friend $p(v_j)$. The edges $\{v_j, p(v_j)\}$ form a spanning tree of the component: each edge points toward an earlier index, and repeated parent steps reach v_1 . Over all components they form a spanning forest with $|P| - c_{\perp}(P)$ edges. The same data therefore certify both a maximal forest and an optimal open-ended sequence.

Exactly one point of each component is omitted from the *selected sequence*. This does not mean that one point of each original component survives in the final residual. A later move in a lower component can delete an omitted point of a higher component. This distinction is important for checking the construction.

4 Graph rank, consequences, and small examples

4.1 A familiar invariant, not an identification of structures

Corollary 4.1 (Graphic-matroid and incidence rank). *For a finite poset P and any field K ,*

$$o_{\perp}(P) = r(G_{\perp}(P)) = \text{rk}_K B,$$

where B is any oriented vertex–edge incidence matrix of $G_{\perp}(P)$. Equivalently, $o_{\perp}(P)$ is the rank of the graphic matroid of $G_{\perp}(P)$.

Proof. The forest characterization of $r(G)$ was established in Lemma 3.2. For the matrix statement, each incidence column has coordinate sum zero within its connected component. Hence the column rank is at most $|V(G)| - c(G)$. The columns belonging to a spanning forest are independent over every field: a leaf row singles out its incident edge, whose coefficient must vanish in any dependence, and leaf removal finishes the induction. The lower bound is therefore the same number. $\square$

In characteristic two, the signs disappear, so an edge column simply has a 1 at each endpoint. This provides an independent, easily implemented algebraic check. The corollary identifies *ranks*; it does not identify the entire tree of friendly sequences with a graphic matroid, and does not assert that every forest orientation gives a legal sequence.

4.2 Monotonicity and one-point changes

Corollary 4.2. *For finite posets the following statements hold.*

- (i) $\mathbf{o}_\perp(P) = \mathbf{o}_\perp(P^{\text{op}})$, and isomorphic incomparability graphs have equal friendly order type.
- (ii) $\mathbf{o}_\perp(P) = 0$ exactly when P is a chain.
- (iii) For nonempty P , $\mathbf{o}_\perp(P) \leq |P| - 1$, with equality exactly when its incomparability graph is connected.
- (iv) Passing to an induced subposet, or adding order comparisons while retaining a partial order, cannot increase friendly order type.

Proof. Duality preserves incomparability, proving (i). Graph rank is zero precisely for an edgeless graph, proving (ii). Statement (iii) is the component formula. For (iv), deleting vertices or edges cannot increase the maximum size of a forest. Adding comparisons deletes incomparability edges. $\square$

Proposition 4.3 (Insertion rule). *Let P' be a finite poset obtained from an induced subposet P by adding one point x . Let t be the number of components of $G_\perp(P)$ that contain at least one friend of x . Then*

$$\mathbf{o}_\perp(P') - \mathbf{o}_\perp(P) = t.$$

Proof. If $t = 0$, the new point is isolated and both the vertex count and component count increase by one. If $t > 0$, the new point merges exactly those t components into one, so $c(G_\perp(P')) = c(G_\perp(P)) - t + 1$. Substitute into Theorem 3.1. $\square$

4.3 Chains, antichains, and equal ordinary invariants

Write C_n for an n -element chain and A_n for an n -element antichain. For $n \geq 1$,

$$\mathbf{o}_\perp(C_n) = 0, \quad \mathbf{o}_\perp(A_n) = n - 1.$$

For nonempty finite A, B , the ordinal sum and disjoint sum satisfy

$$\mathbf{o}_\perp(A + B) = \mathbf{o}_\perp(A) + \mathbf{o}_\perp(B), \quad \mathbf{o}_\perp(A \sqcup B) = |A| + |B| - 1.$$

The first follows because there are no cross-block incomparability edges; the second because every cross-block pair is incomparable, making the graph connected. These two special operation rules are consistent with Vialard's existing binary-sum formulas; they are not claimed here as new binary-sum results [1, Proposition 4.1].

Example 4.4 (A four-point separation). Let

$$P = A_2 + A_2, \quad Q = C_2 \sqcup C_2.$$

Both have four elements, height two, ordinary width two, and maximal order type four. The incomparability graph of P consists of two disjoint edges, whereas the graph of Q is the connected graph $K_{2,2}$. Thus

$$\mathbf{o}_\perp(P) = 2, \quad \mathbf{o}_\perp(Q) = 3,$$

and Theorem 2.2 gives

$$\mathbf{w}(M^r(P)) = \omega^2, \quad \mathbf{w}(M^r(Q)) = \omega^3.$$

This directly demonstrates, already on finite bases, that the usual three ordinal invariants of a base do not determine its multiset-order width.

For every fixed $n \geq 1$, every integer r with $0 \leq r \leq n-1$ occurs as $\mathbf{o}_\perp(P)$ on an n -element poset. One realization is

$$P = A_{r+1} + C_{n-r-1},$$

where a zero-sized final chain is omitted. Consequently the possible multiset-order widths of n -element bases are exactly $1, \omega, \dots, \omega^{n-1}$.

5 Finite lexicographic substitutions

Definition 5.1 (Substitution). Let R be a finite poset and let P_i be a nonempty finite poset for every $i \in R$. The substitution $R[P_i : i \in R]$ has elements (i, x) with $x \in P_i$. Within a fiber it uses the order of P_i , and between distinct fibers it declares every point of P_i below every point of P_j exactly when $i <_R j$.

Let $C(R)$ denote the set of incomparability components of R . A singleton component is an isolated vertex of $G_\perp(R)$.

Theorem 5.2 (Substitution formula). *Under the preceding hypotheses,*

$$\mathbf{o}_\perp(R[P_i : i \in R]) = \sum_{\substack{D \in C(R) \\ |D| > 1}} \left(\sum_{i \in D} |P_i| - 1 \right) + \sum_{\{i\} \in C(R)} \mathbf{o}_\perp(P_i). \quad (5)$$

All sums in this formula are finite sums of integers.

Proof. Whenever $i \perp_R j$, all vertices of the two fibers are mutually joined by cross-fiber incomparability edges. Suppose D is a nontrivial connected component of $G_\perp(R)$. Every vertex i of D has a neighbor in D . Two vertices in the same fiber over i can therefore be joined by a length-two path through any vertex of a neighboring fiber. Vertices in different fibers over D can be joined by lifting a path in D . Hence the union of the fibers over D forms one connected component.

There are no edges between fibers lying over distinct index components. For an isolated index i , the fiber retains exactly its own internal components. Thus a nontrivial D contributes $\sum_{i \in D} |P_i| - 1$ to graph rank, and an isolated index contributes $|P_i| - c_\perp(P_i) = \mathbf{o}_\perp(P_i)$. Apply Theorem 3.1. $\square$

Corollary 5.3 (Constant fiber). *Suppose every fiber is the same nonempty finite poset P . If R has s isolated vertices and q nontrivial incomparability components, then*

$$c_\perp(R[P]) = q + sc_\perp(P), \quad \mathbf{o}_\perp(R[P]) = |R||P| - q - sc_\perp(P).$$

This provides a genuinely structural compositional rule: an isolated index requires the internal friendly order type of its fiber, while a nontrivial index component forgets the fiber orders and retains only their sizes. Empty fibers are excluded because they can destroy connecting paths; they can be handled by first deleting their indices and applying the theorem to the remaining index poset. An empty index yields the empty poset.

6 Cartesian products: only the endpoints split off

The Cartesian order is coordinatewise: $(a, b) \leq (a', b')$ exactly when $a \leq a'$ and $b \leq b'$. The next structural lemma applies even without finiteness.

Lemma 6.1 (No middle ordinal cut in a nontrivial product). *Let A, B be posets, each with at least two elements. There is no partition $A \times B = L \dot{\cup} U$ with $|L|, |U| \geq 2$ and every point of L strictly below every point of U .*

Proof. Suppose there is such a partition. Let $L_A, U_A \subseteq A$ and $L_B, U_B \subseteq B$ be its coordinate projections. Every point of L_A is at most every point of U_A , and similarly in B . It follows by antisymmetry that

$$|L_A \cap U_A| \leq 1, \quad |L_B \cap U_B| \leq 1.$$

Also, $L_A \cup U_A = A$ and $L_B \cup U_B = B$.

If $a \in L_A \setminus U_A$, its entire row $\{a\} \times B$ lies in L . Therefore there can be no $b \in U_B \setminus L_B$, whose entire column would lie in U . We have proved

$$L_A \setminus U_A \neq \emptyset \implies U_B \subseteq L_B.$$

Reversing L, U yields the analogous implication $U_A \setminus L_A \neq \emptyset \implies L_B \subseteq U_B$.

Since $|L| \geq 2$, one of its projections has at least two points; exchange coordinates if necessary so that $|L_A| \geq 2$. Then $L_A \setminus U_A \neq \emptyset$, so $U_B \subseteq L_B$, making U_B a singleton. Since $|U| \geq 2$, we now have $|U_A| \geq 2$. Hence $U_A \setminus L_A \neq \emptyset$, and $L_B \subseteq U_B$. Consequently $B = L_B \cup U_B$ is a singleton, a contradiction. $\square$

Theorem 6.2 (Incomparability graph of a product). *If $|A|, |B| \geq 2$, the incomparability graph of $A \times B$ has exactly one nontrivial connected component. Its only possible additional components are the singleton least element and the singleton greatest element of the product, when they exist.*

Proof. The product is not a chain. If either factor has an incomparable pair, fixing the other coordinate gives one in the product. Otherwise take $a_0 < a_1$ and $b_0 < b_1$; then $(a_0, b_1) \perp (a_1, b_0)$. Thus there is a nontrivial component C .

By Lemma 3.3, all other components are linearly ordered below or above C . Let L be the union of the components below C . If $|L| \geq 2$, then L and its complement give a forbidden cut from Lemma 6.1, since the complement contains C . Thus $|L| \leq 1$. Similarly the union of components above C has at most one point. If the lower point exists, it is the least point of the product; if the upper point exists, it is the greatest. Conversely a least or greatest point is comparable to every point and therefore isolated in the graph. $\square$

For a nonempty poset A , let $b(A), t(A) \in \{0, 1\}$ indicate the existence of a least and greatest point, respectively.

Corollary 6.3 (Finite Cartesian formula). *For finite A, B with $|A|, |B| \geq 2$,*

$$\boxed{o_{\perp}(A \times B) = |A||B| - 1 - b(A)b(B) - t(A)t(B).} \quad (6)$$

Consequently

$$w(M^r(A \times B)) = \omega^{|A||B| - 1 - b(A)b(B) - t(A)t(B)}.$$

Proof. The product has a least point exactly when both factors have least points, and likewise for greatest points. The graph therefore has $1 + b(A)b(B) + t(A)t(B)$ components. Apply Theorems 3.1 and 2.2. $\square$

If either factor is empty, the product is empty and has friendly order type zero. If one factor has one point, the product is isomorphic to the other factor. These exceptions must be handled before using (6).

Corollary 6.4 (Several factors and finite grids). *For $d \geq 2$ finite factors $P_1, \dots, P_d$, each of size at least two,*

$$o_{\perp}\left(\prod_{i=1}^d P_i\right) = \prod_{i=1}^d |P_i| - 1 - \prod_{i=1}^d b(P_i) - \prod_{i=1}^d t(P_i).$$

In particular, for $m_i \geq 2$,

$$o_{\perp}(C_{m_1} \times \dots \times C_{m_d}) = \prod_{i=1}^d m_i - 3.$$

For the Boolean lattice $B_d = C_2^d$ with $d \geq 2$,

$$o_{\perp}(B_d) = 2^d - 3, \quad w(M^r(B_d)) = \omega^{2^d - 3}.$$

The finite product formula is notably simpler than a general computation of ordinary width: away from the empty and singleton exceptions, only cardinalities and endpoint indicators are needed. For example, $o_{\perp}(C_2 \times C_3) = 3$ and $o_{\perp}(B_3) = 5$.

7 Well-ordered sums and a transfinite evaluation theorem

7.1 Additivity beyond binary sums

For an ordinal δ and wpos P_i ($i < \delta$), their ordinal sum $P = \sum_{i < \delta} P_i$ puts every point of an earlier block below every point of a later block. Empty blocks are permitted. This sum is a wpo: in an infinite bad sequence the block indices would have to be nonincreasing, hence eventually constant, giving an infinite bad sequence in one block.

The sums of ordinals below are *ordinary ordinal sums*, not natural sums and not commutative sums. Set

$$S_0 = 0, \quad S_{i+1} = S_i + o_{\perp}(P_i), \quad S_{\lambda} = \sup_{i < \lambda} S_i \quad (\lambda \text{ a limit}).$$

Theorem 7.1 (Well-ordered-sum additivity). *For every ordinal δ and family of wpos $(P_i)_{i < \delta}$,*

$$o_{\perp}\left(\sum_{i < \delta} P_i\right) = \sum_{i < \delta} o_{\perp}(P_i). \quad (7)$$

Proof. We prove the assertion for all such representations simultaneously, by transfinite induction on $o(P)$, where $P = \sum_{i < \delta} P_i$. If P is empty, both sides vanish. A point $x \in P_i$ has an incomparable friend in P exactly when it has one in P_i . Moreover,

$$P_{\not\leq x} = \left(\sum_{j < i} P_j\right) + (P_i)_{\not\leq x}. \quad (8)$$

Indeed, all later blocks are deleted, all earlier blocks survive, and only the upset of x is deleted in its own block.

Equation (1) gives $o(P_{\not\leq x}) < o(P)$, so the induction hypothesis applies to the representation on the right of (8). It yields

$$o_{\perp}(P_{\not\leq x}) = S_i + o_{\perp}((P_i)_{\not\leq x}).$$

Substitution into (2) gives

$$\begin{aligned} o_{\perp}(P) &= \sup_{\substack{i < \delta \\ x \in P_i, (P_i)_{\perp x} \neq \emptyset}} (S_i + o_{\perp}((P_i)_{\perp x}) + 1) \\ &= \sup_{\substack{i < \delta \\ o_{\perp}(P_i) > 0}} (S_i + o_{\perp}(P_i)). \end{aligned}$$

The second equality uses associativity and continuity of ordinal addition in its right argument, together with (2) within P_i . Blocks with friendly order type zero permit no move and contribute no increase to the partial sums. The displayed supremum is therefore exactly $S_{\delta} = \sum_{i < \delta} o_{\perp}(P_i)$, including the case where all blocks have value zero. This completes the induction. $\square$

The binary case is already a published fact [2, Theorem 5.4.8(1)]. The proof above records the arbitrary well-ordered version and, in particular, makes the limit step explicit. No assumption of countability, a computable presentation, or a bound on component sizes is imposed.

7.2 All wpos with finite incomparability components

Lemma 7.2. *The incomparability components of a wpo, ordered as in Lemma 3.3, form a well-order.*

Proof. Take any nonempty collection of components. The union of their points is a nonempty subposet of a wpo, hence has a minimal point x . The component containing x is least in the collection: a lower component would supply a point strictly below x . Thus every nonempty collection has a least component. $\square$

Theorem 7.3 (Finite-component transfinite formula). *Let P be a wpo whose incomparability components are all finite. Enumerate them in order as $(C_i)_{i < \delta}$. Then*

$$o_{\perp}(P) = \sum_{i < \delta} (|C_i| - 1), \quad w(M^r(P)) = \omega^{\sum_{i < \delta} (|C_i| - 1)}. \quad (9)$$

More generally, if $P = \sum_{i < \delta} Q_i$ with arbitrary finite blocks, then

$$o_{\perp}(P) = \sum_{i < \delta} (|Q_i| - c_{\perp}(Q_i)).$$

Proof. Each C_i has connected incomparability graph, so its friendly order type is $|C_i| - 1$. Apply Theorem 7.1, followed by Theorem 2.2. For arbitrary finite blocks apply Theorem 3.1 to each block first. $\square$

This is not the ambiguous expression “ $|P|$ minus the number of components” for an infinite cardinal $|P|$. It is an ordinal sum of explicit finite ranks, in the actual order of the components. Its order sensitivity is essential.

7.3 Ordinal examples and realizability

If Q is finite and $r = o_{\perp}(Q)$, then for every ordinal α ,

$$o_{\perp}\left(\sum_{i < \alpha} Q\right) = r \cdot \alpha. \quad (10)$$

The finite number r is the *left* factor in this ordinary ordinal product. For $r = 2$ and $\alpha = \omega + 3$, the answer is $\omega + 6$, not $\omega \cdot 2 + 6$.

Example 7.4 (A transfinite sum of Boolean lattices). Let $Q = B_3$, so $r = 5$, and take $\alpha = \omega^2 + \omega \cdot 3 + 5$. Then

$$o_{\perp} \left(\sum_{i < \alpha} B_3 \right) = 5 \cdot \alpha = \omega^2 + \omega \cdot 3 + 25.$$

Its multiset-order width is

$$\omega^{\omega^2 + \omega \cdot 3 + 25}.$$

This base has infinitely many finite incomparability components, including the isolated endpoints inside each Boolean-lattice block. They contribute zero locally; the connected six-point middle contributes five.

Corollary 7.5 (Every ordinal is a friendly order type). *For every ordinal α , the wpo $P_{\alpha} = \sum_{i < \alpha} A_2$ satisfies*

$$o_{\perp}(P_{\alpha}) = \alpha, \quad w(M^r(P_{\alpha})) = \omega^{\alpha}.$$

This applies to uncountable ordinals as well as countable ones. It is an existence statement about set-sized orders, not an effective notation system for arbitrary ordinals. We regard it as an immediate corollary, not as a separate priority claim.

8 Why the infinite generalization needs more information

The finite graph formula might suggest two stronger extrapolations: that friendly order type is always determined by the abstract incomparability graph, or that a connected wpo has maximal order type with a fixed one-point correction. Both extrapolations fail. We give direct residual calculations, so the examples do not depend on an ordinal-subtraction convention.

For ordinals α, β , let

$$g(\alpha, \beta) = o_{\perp}(\alpha \sqcup \beta),$$

where each ordinal is regarded as a chain. If either coordinate is zero, $g(\alpha, \beta) = 0$. If both are positive, every point has a friend in the other chain and

$$g(\alpha, \beta) = \sup (\{g(\xi, \beta) + 1 : \xi < \alpha\} \cup \{g(\alpha, \eta) + 1 : \eta < \beta\}). \quad (11)$$

In particular, induction gives

$$g(m, n) = m + n - 1 \quad (m, n \geq 1 \text{ finite}), \quad (12)$$

$$g(\alpha, 1) = \alpha \quad (\alpha > 0), \quad (13)$$

$$g(\omega, n) = \omega + (n - 1) \quad (n \geq 1), \quad (14)$$

$$g(\omega, \omega) = \omega \cdot 2, \quad (15)$$

$$g(\omega + 1, n) = \omega + n \quad (n \geq 1), \quad (16)$$

$$g(\omega + 1, \omega) = \omega \cdot 2 + 1. \quad (17)$$

Here is the verification of the steps involving limits. For (14), reducing the infinite coordinate gives finite values cofinal in ω , and reducing the finite coordinate gives the successors $\omega + 1, \dots, \omega + (n - 1)$. For (15), the values from (14), followed by $+1$, have supremum $\omega \cdot 2$. For (16), reducing $\omega + 1$ to ω gives the value $\omega + n$, and all other branches are bounded by it. Finally, in (17), reducing the first coordinate to ω gives $g(\omega, \omega) + 1 = \omega \cdot 2 + 1$, while reducing either coordinate to a finite ordinal contributes values with supremum only $\omega \cdot 2$.

Proposition 8.1 (Abstract graph invariance fails in the infinite case). *The countable wpos*

$$X = \omega \sqcup \omega, \quad Y = (\omega + 1) \sqcup \omega$$

have isomorphic incomparability graphs but different friendly order types:

$$G_{\perp}(X) \cong G_{\perp}(Y) \cong K_{\aleph_0, \aleph_0}, \quad o_{\perp}(X) = \omega \cdot 2 \neq \omega \cdot 2 + 1 = o_{\perp}(Y).$$

Proof. Each side of either disjoint sum is a countably infinite chain; all and only cross-chain pairs are incomparable. The graph assertion follows. The rank values are (15) and (17). $\square$

Proposition 8.2 (Connectivity and maximal order type do not suffice). *Let $Z = (\omega \cdot 2) \sqcup 1$ and $Y = (\omega + 1) \sqcup \omega$. Both incomparability graphs are connected, and*

$$o(Z) = o(Y) = \omega \cdot 2 + 1,$$

but

$$o_{\perp}(Z) = \omega \cdot 2, \quad o_{\perp}(Y) = \omega \cdot 2 + 1.$$

Proof. Connectivity is immediate from the cross-chain edges. The friendly values follow from (13) and (17). For clarity, the maximal-order-type values can also be checked from bad-tree ranks. Put $h(\alpha, \beta) = o(\alpha \sqcup \beta)$. It has the same recursion as (11), but boundary values $h(\alpha, 0) = \alpha$ and $h(0, \beta) = \beta$. Induction gives $h(\alpha, 1) = \alpha + 1$, $h(\omega, n) = \omega + n$, $h(\omega, \omega) = \omega \cdot 2$, and $h(\omega + 1, n) = \omega + n + 1$. At $(\omega + 1, \omega)$, the branch through (ω, ω) contributes $\omega \cdot 2 + 1$; the remaining cofinal branches have supremum $\omega \cdot 2$. Thus $h(\omega + 1, \omega) = \omega \cdot 2 + 1$ as claimed. $\square$

These examples put a precise limit on the proposed method. Decomposition into ordered incomparability components still works for every wpo, and Theorem 7.1 reduces the problem to individual components. What fails is the replacement of an arbitrary infinite component by a finite-style graph rank or a universal predecessor operation on its maximal order type. Its internal well-founded order structure can matter.

9 Algorithms and reproducible verification

9.1 Value computation and witness construction

If the incomparability graph is supplied as an adjacency list with n vertices and m edges, graph traversal computes its number of components and hence $o_{\perp}(P)$ in $O(n+m)$ graph operations. If a full comparison matrix is supplied, building the incomparability graph takes $O(n^2)$ time. If only Hasse-cover relations are supplied, transitive closure must be computed first; a straightforward implementation uses $O(n^3)$ time. The graph-only value algorithm does not require a particular orientation of the comparable pairs, but construction of a legal witness does.

The constructive proof supplies a polynomial-time witness algorithm: find the ordered components, build a connected-prefix linear extension in each, and read all but the first point of each extension in reverse order, starting from the highest component. A direct comparison-matrix implementation takes at most $O(n^3)$ time. More careful maintenance of minimal candidates improves this bound, but is not needed for the claim.

The reference exact-rank computation is deliberately separate:

```
@cache
def rank(mask):
    answer = 0
    for x in vertices(mask):
        if incomparable[x] & mask:
            next_mask = mask & ~upset[x]
            answer = max(answer, 1 + rank(next_mask))
    return answer
```

It uses neither connected components nor the proved formula. At most 2^n residual masks are examined, with at most n candidate moves at each mask: $O(n2^n)$ bit-mask operations and $O(2^n)$ stored ranks, before accounting for the bit length of the masks. This is an exponential audit algorithm, not the recommended value algorithm for large finite inputs.

The supplied implementation uses arbitrary-precision integer masks. The complexity bounds just stated count graph/comparison operations or abstract bit-mask operations; they should not be confused with constant-time bounds on arbitrarily large Python integers.

9.2 Exhaustive finite checks

A poset on integer labels $0, \dots, n-1$ is *naturally labelled* here when $x <_P y$ implies $x < y$ as integers. The generator adds the largest label as a new maximal point, choosing any order ideal of the previous poset as its predecessor set. This constructs every naturally labelled relation exactly once: deleting the largest label uniquely recovers the earlier poset and the chosen ideal.

Every finite poset admits a linear extension, so every isomorphism type has at least one naturally labelled representative. The enumeration is not an isomorph-free enumeration and is not a count of all arbitrary labellings. This distinction is relevant to interpreting the following actual run.

n	Naturally labelled relations	Exact-rank comparisons	Result
0	1	1	pass
1	1	1	pass
2	2	2	pass
3	7	7	pass
4	40	40	pass
5	357	357	pass
6	4,824	4,824	pass
7	96,428	96,428	pass
Total	101,660	101,660	pass

For every listed relation the audit compared the exact residual rank with $n - c$, computed the incidence-matrix rank over $\mathbb{F}_2$, constructed an optimal witness, and checked each selected point and friend against the actual residual. A further 351 seeded relabelling tests checked that the implementation does not accidentally depend on natural labels.

For additional reproducibility, the distributions by friendly order type are stored in `data/rank_distributions.csv`. For example, at $n = 7$ the numbers with friendly ranks $0, 1, \dots, 6$ are

$$1, 6, 30, 160, 1047, 8906, 86278,$$

respectively. These are outputs of the supplied enumeration, not asserted new integer sequences or independent proof certificates for infinite cases.

9.3 Product and substitution checks

The product audit tested all ordered pairs of naturally labelled factors of sizes two through five. There are $2 + 7 + 40 + 357 = 406$ such factor relations, giving $406^2 = 164,836$ product cases. For each case it compared the directly computed number of product incomparability components with Theorem 6.2. The 241 cases with product size at most nine were also checked by the independent exact-rank recurrence. The other product cases were *graph-structure checks*, not exponential exact-rank checks on up to 25 points.

The substitution audit used every naturally labelled index of sizes zero through three and every assignment of nonempty naturally labelled fibers of sizes one through three. There are ten available nonempty fiber relations. The total is

$$1 + 1 \cdot 10 + 2 \cdot 10^2 + 7 \cdot 10^3 = 7,211.$$

All 7,211 cases, each with at most nine points, passed both the exact-rank comparison and the direct graph-rank comparison.

The recorded run used Python 3.13.5 and completed all audits in about 31 seconds in the execution environment used for this report. This timing is not a portable benchmark or an asymptotic performance claim. Exact counts, individual timings, and outputs are in `data/audit_results.json`.

9.4 How to reproduce the package

The code needs only the Python standard library and is written for Python 3.10 or later. From the package root run

```
python3 code/audit.py --max-n 7 --output data
```

The check through eight elements is optional and substantially slower:

```
python3 code/audit.py --max-n 8 --output data_n8
```

No eight-element exhaustive run is claimed in this article. The finite proof covers all sizes independently of these computations.

A simple library use is

```
from friendly_order import chain, cartesian

P = cartesian(chain(2), chain(3))
assert P.friendly_formula() == 3
assert P.exact_friendly_rank() == 3
certificate = P.optimal_witness()
P.verify_witness(certificate)
```

Run this with `code/` on the Python module search path, for example from inside that directory. The data file `examples.json` contains explicit upset masks, component extensions, omitted roots, and friend certificates for the examples in this article.

10 Research status, dependencies, and remaining questions

10.1 What has and has not been established

The finite component theorem is a universal statement with a complete constructive proof. The substitution and Cartesian formulas are consequences of proved graph-structure lemmas. The well-ordered-sum formula is proved by transfinite induction, not extrapolated from finite tests. The countable counterexamples are verified by explicit ordinal recurrences.

The multiset-width conclusions have a different dependency: they apply Vialard’s published Theorem 2.2. The present note does not replace its general proof. Nor does it solve the broader problem of computing friendly order type on arbitrary infinite incomparability-connected wpos. The finite-component class is a significant exact domain, but many infinite wpos have infinite components and lie outside it.

The relation to other ordinal-invariant work is consistent with the perspective of the finitary-powerset paper of Abriola and collaborators, which emphasizes that additional invariants or restricted structural classes may be needed to regain compositional computation [3]. Here, for finite bases, the additional information is an elementary graph component count. In infinite examples even the abstract graph can lose the necessary order information.

10.2 Priority and literature check

The source audit examined Vialard’s published MFCS 2023 paper, the author’s 2024 thesis copy, and the finitary-powerset follow-up. Searches included “friendly order type”, “friendly order type finite posets”, and combinations with “components” and “incomparability”. The published conclusion and the thesis conclusion explicitly present further study of friendly order type as a research direction. No finite component formula was located in the materials inspected.

This establishes the provenance of the question and describes the search, not a proof of novelty. The formula may occur elsewhere, be known informally, or be obtainable from a result not found in that search. The appropriate status is *a proved result in this unrefereed manuscript, with priority unverified*. The source lookup date was 19 September 2026.

10.3 Precise directions left by the result

The reduction to incomparability components suggests a focused remaining problem: classify $\mathbf{o}_\perp(C)$ for infinite incomparability-connected wpos C . Propositions 8.1 and 8.2 show that both the abstract graph and maximal order type alone can be insufficient. A useful next invariant would have to retain some interaction between finite terminal structure and infinite bad-sequence ranks.

Another concrete question concerns *which* subsets of a finite poset can occur as the selected support of an open-ended bad sequence. The rank theorem determines the maximum support size and constructs an optimum, but does not characterize all feasible supports or identify their family with the independent sets of a matroid. Confusing equality of maximal ranks with an equality of combinatorial structures would be an unjustified extension.

Finally, efficient incremental witness maintenance can be studied alongside Proposition 4.3. The numerical rank change under a valid one-point insertion is completely determined by the old components met by the new point. Updating a particular connected linear extension while retaining a short certificate is a more detailed algorithmic question.

11 Conclusion

For finite posets, the residual definition of friendly order type conceals a simple graph invariant:

$$o_{\perp}(P) = r(G_{\perp}(P)) = |P| - c_{\perp}(P).$$

The upper bound is a forest-rank potential, and the lower bound is a connected-prefix linear-extension construction. Their combination gives an exact value, an optimal sequence, and a spanning-forest certificate.

This answers the finite structural subproblem of the published program and supplies further compositional formulas, notably

$$o_{\perp}(A \times B) = |A||B| - 1 - b(A)b(B) - t(A)t(B) \quad (|A|, |B| \geq 2).$$

Well-ordered additivity extends the evaluation to all wpos with finite incomparability components, while explicit infinite examples explain why additional structure is needed beyond that class. The reproducible checks support the implementation and finite statements; the proofs, not the checks, establish the general conclusions.

A A compact proof certificate for the finite theorem

The complete finite theorem can be checked against the following four claims, each proved in the body of the article.

1. A legal move deletes x but not a friend y . A spanning forest of the residual can be extended by the edge xy , so graph rank drops by at least one.
2. Incomparability components are linearly ordered ordinal-sum blocks. The direction of comparison with an outside vertex cannot change along an incomparability edge.
3. A connected finite component has a linear extension with connected prefixes. Otherwise some nonempty proper ideal would be below every remaining point and disconnect the graph.
4. Reverse each such extension, omit its first point, and process blocks from high to low. Every selected point has an earlier-prefix friend. The result has exactly one fewer selected point than vertices in each component.

The first claim proves the bound $o_{\perp}(P) \leq |P| - c_{\perp}(P)$, and the remaining claims produce a witness attaining it. None assumes that every legal move preserves connectivity, or that the omitted points must remain in the final residual.

B Artifact manifest

File	Purpose
<code>article.tex</code>	Complete source of this article; bibliography embedded.
<code>article.pdf</code>	Compiled article.
<code>README.md</code>	Build, reproduction, scope, and status instructions.
<code>build.sh</code>	Two-pass PDF build using <code>pdflatex</code> .
<code>code/friendly_order.py</code>	Validated finite-poset representation, exact rank, graph formula, operations, and witnesses.
<code>code/audit.py</code>	Exhaustive and compositional verification driver.
<code>data/audit_results.json</code>	Actual audit counts, timings, and pass status.
<code>data/audit_console.txt</code>	Captured console output of the reported run.
<code>data/rank_distributions.csv</code>	Counts by size and friendly rank.
<code>data/examples.json</code>	Explicit finite examples and certificates.
<code>RESEARCH_STATUS.md</code>	Claim/dependency boundary and source audit.

References

- [1] Isa Vialard. Ordinal Measures of the Set of Finite Multisets. In *48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023)*, LIPIcs **272**, article 87, 87:1–87:15, 2023. DOI: [10.4230/LIPIcs.MFCS.2023.87](https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2023.87). <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2023.87>. In particular, Definition 3.3, Theorem 3.4, Proposition 4.1, and the conclusion. Published version, rather than the early arXiv terminology.
- [2] Isa Vialard. *Measuring Well Quasi-Orders and Complexity of Verification*. Doctoral thesis, Université Paris-Saclay, 2024. Author-hosted copy: <https://isavialard.github.io/home/mwqo.pdf>. In particular, Sections 5.3–5.4 and the conclusion, printed pp. 109–110. The page references here concern this 124-page author copy.
- [3] Sergio Abriola, Simon Halfon, Aliaume Lopez, Sylvain Schmitz, Philippe Schnoebelen, and Isa Vialard. Measuring well-quasi-ordered finitary powersets. arXiv:2312.14587, version 2, 17 July 2024. <https://arxiv.org/abs/2312.14587>. Used for research context, not as a dependency of the component formula proved here.